SEEK THE NATURE OF THE UNIVERSE

> I don't know the meaning of life. I don't think we can answer that question well...**yet**.[68]

We must expand humanity and consciousness to the point where we *are* able to answer that question.[69]

It may be uncomfortable for a lot of people, but I think this is a rational and logical philosophy. Basically saying, "We don't know what the answer is, but let's try to find out."[70]

I think we as humans collectively know part of the answer, but only a tiny part at this point. If we can make civilization last for a million years, we will probably know much more of the answer.[71]

If we date civilization from the first written language, it was about five thousand years ago. This is practically no time at all. If civilization were to last a million more years, we're barely even at the beginning. Future humans will think of us as ancient ancients, like cavemen.[72]

> This is the foundation of my philosophy: I am curious about the nature of the universe.

We can see the archaeological evidence in the fossil record, based on what we know of physics, how we came to be at this point on Earth step by step. But that doesn't explain how the universe came to exist in the first place.[73]

How do a bunch of molecules develop consciousness and feelings? From a physics standpoint, the chain of events from the beginning of the universe

to now is quite well understood. There was a bunch of hydrogen gas that turned into complex molecules and now an assemblage of complex molecules—us humans who can feel, talk, and think. Apparently if you leave enough hydrogen gas sitting around long enough, it starts talking to itself. That's basically what happened here.[74]

Assuming you believe the physics, which appear to be true, the universe started off as quarks and electrons and quickly became hydrogen, helium, lithium, most of the periodic table. Mostly hydrogen, though. Then, over a long period of time—13.8 billion years—that hydrogen became sentient.[75]

Where along the path from a bunch of hydrogen to human beings did consciousness start? It's crazy.[76] I wonder, is everything conscious or is nothing conscious? Maybe it's just degrees of consciousness or concentrations of consciousness.[77]

Total collective consciousness is how many people there are times the average amount of consciousness per person. If we can expand consciousness by creating more humans and more digital intelligence, then our opportunity to understand the meaning of life is that much greater. I would call this the philosophy of curiosity—to better understand the nature of reality.[78]

What I think Douglas Adams was saying in his book *The Hitchhiker's Guide to the Galaxy* is "The universe is the answer." We need to figure out what questions to ask about the answer that is the universe. The question is the hard part, and if we can properly frame the question then the answer, relatively speaking, is easy.

We need to expand the scope and scale of consciousness in order to understand what questions to ask of the universe. This is the path forward. If we do, we will be better able to understand the nature of the universe and understand the meaning of life. As just one example, we have to get back to the moon and build a science base there. I think we could learn a lot about the nature of the universe.[79]

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THINK LIKE A PHYSICIST

I try to be hyperrational.

If the reasoning fits, and you're not violating the laws of physics, that's the thing you should try to do.

These things just don't seem that crazy to me.[80]

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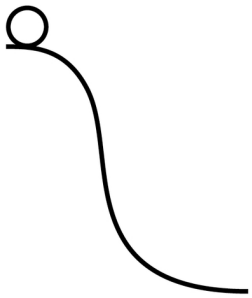
OBSESS OVER TRUTH

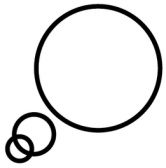
> Start somewhere. Then be prepared to question your assumptions, fix what you did wrong, and adapt to reality.[81]

I am obsessed with truth. Obsessed.[82] If you're going to come up with a good solution, the truth is really, really important.[83]

This obsession with truth is why I studied physics, because physics attempts to understand the truth of the universe. Physics is about finding the provable truths of the universe, and finding truths that have predictive power.[84]



PHYSICS
IS LAW.



EVERYTHING ELSE IS
A RECOMMENDATION.

> Physics is law. Everything else is a recommendation. I've met many people who can break the laws of man, but I have never met anyone who could break the laws of physics.

For me, physics was a natural thing to study. It was intrinsically interesting to understand the nature of the universe. I also studied computer science and information theory to understand logic. (Information theory studies the transmission, processing, extraction, and use of information.) There's an

argument that information theory is actually operating at a more fundamental level than physics.[85]

> Truth matters to me a lot. Pathologically, it matters to me.[86]

No matter how smart you are, you will make some number of mistakes. Everyone makes mistakes. It's just a question of how many and how often.[87]

In business and personal life, wishful thinking causes a lot of mistakes. You have to ask whether something is true or not. If something ever feels too easy or doesn't quite make sense…it is probably wishful thinking.[88]

Wishful thinking is a natural human tendency. It's a challenge to tell the difference between believing in a new idea and persevering or pursuing an unrealistic dream. You need to be rigorous in your self-analysis. Focus on something you're confident will have high value to someone else. Be rigorous in making that assessment too.[89]

The real test of any startup is how well it responds to adversity and adapts. When most things start out they don't make much sense, but as long as you adapt quickly you can make the company work.

> Being tenacious and super focused on the truth is extremely important. Look for feedback from all sources.[90]
>
> If you have beliefs that are incompatible with a rocket getting to orbit, the rocket will not get to orbit. Physics is a harsh judge.[91]

Do you have the right fundamental axioms, or truths? Are they relevant? Are you making the right conclusions based on those truths? That's the essence of critical thinking, and yet it is amazing how often people fail to do that. Wishful thinking is innate in the human brain. You want things to be the way you wish them to be, so you tend to filter out information you shouldn't.[92]

That's why I always assume we're losing, even when it looks like we might win.[93]

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FIRST-PRINCIPLES THINKING

Q: Where does first-principles thinking come from?

When you want to do something new, you have to apply the physics approach. Physicists discover counterintuitive new things, like quantum mechanics. They do that by thinking from "first principles": building their reasoning from the ground up.[94]

> I would encourage people to use the mental tools of physics and apply them broadly in life. They are the best tools.[95]

The normal way we conduct our lives is reasoning by analogy. That means we do something because it's similar to something else, or what other people are doing.

When you think this way, you only get slight iterations. It's easier to reason by analogy rather than from first principles, so that's what we do most of the time.[96] And in most of life, we should reason by analogy. Otherwise, mentally, you wouldn't be able to get through the day. It would be too much thinking.[97]

But for important things, that kind of thinking is too bound by convention or prior experiences. You will hear, "It's always been done this way," or "Nobody's ever done it." That is a ridiculous way to think.

Don't just follow the trends. You can avoid following trends by thinking with the physics approach, first principles. It's a powerful, powerful method for life in general.

Look at the fundamentals and construct your reasoning from there. Then see if you have a conclusion that works or doesn't work.[98] It might or might not be different from what people have done in the past.[99]

Q: How do you apply first-principles thinking?

Break something down to the most fundamental principles. Start by asking: What am I most confident is true at a foundational level? That sets your axiomatic base. Then you reason up from there. Then you check your conclusions against the axiomatic truths.

For instance, to approach any new technology problem, make sure you're not violating physics with a first-principles analysis. A basic question in physics would be: Am I violating conservation of energy or momentum? If so, it's not going to work. That's just to establish if this idea is possible.[100]

It's hard to think this way. It takes a lot of effort. But if you're trying to do something new, it's the best way to think.[101]

Q: How have you applied first-principles thinking to building companies?

Here's an example from early in building Tesla. People said battery packs were too expensive to make cheap electric cars. They assumed they would always be expensive, because they had been in the past. That's pretty dumb. If you applied that reasoning to everything new, then you would never try anything new. "Oh, nobody wants a car. Horses are great; we're used to them. They can eat grass. There's lots of grass all over the place. There's no gasoline available. So people will never buy gas cars." People did say that, a lot.

People assumed batteries for electric vehicles would always cost $600 per kilowatt hour. The first-principles approach to battery costs is this: What are the batteries made of? What are the materials that make up the batteries? What is the market value of those material constituents?

It's got cobalt, nickel, aluminum, carbon, some polymers for separation, and a steel can. Okay, what if we bought that amount of material at the London

Metal Exchange? What would each of those things cost? Oh, geez, it's only $80 per kilowatt hour. So clearly, we just need to think of clever ways to take those materials and combine them into the shape of a battery cell. That's how I knew it was possible to build batteries much much cheaper than anyone else realized.[102]

> Great differences in technology exist in the world, which even hardcore technologists are unaware of.
>
> What is simple in one arena is often profound in another.[103]

The first-principles approach is a good way to understand what new things are possible. It doesn't mean you'll be successful, but at least you can determine if success is one of the possibilities, and that is important. This is how I decided to start SpaceX.[104]

First-principles thinking built SpaceX. Most people think, "Historically, all rockets have been expensive. Therefore, in the future, all rockets will be expensive." But that's not true.[105] This is where it's helpful to use the analytical approach again.[106]

The way we applied first-principles thinking to rocketry was asking, "What are the materials that go into a rocket?"[107] A rocket is made from aluminum, titanium, copper, and carbon fiber. Break it down further and ask, "How much of each material is used? Now, what is the cost of all these raw components?"[108]

If you have them stacked on the floor and could wave a magic wand to create the rocket, what would the cost of the rocket be? We imagine the cost of rearranging the atoms was zero.[109]

That's going to set the floor of the cost of the rocket. I call this the "magic wand number," the hypothetical best-case scenario. For rockets, that turned

out to be a relatively small number, well under 5 percent of the current cost, in some cases closer to 1 or 2 percent.[110] The manufacturing must be very inefficient if the raw material cost is only 1 or 2 percent of the finished product.[111]

I was able to see a great deal of room for improvement.[112] Now our challenge was to figure out how to get the atoms in the right shape more efficiently.[113]

That first-principles thought process around the rocket became general purpose for all parts. I call it "The Idiot Index." How much more does a finished product cost than the cost of its materials? If a part or product had a high Idiot Index, we could cut the cost with more efficient manufacturing techniques.[114]

A component that costs $1,000 when the aluminum it was made of costs only ten dollars likely has a design that is too complex or an inefficient manufacturing process. If the ratio is high, you're an idiot.

One part of the rocket, the half nozzle jacket, cost $13,000. But it was only made of $200 worth of steel. I expect all my engineers to know all the best and worst parts in their systems as judged by the idiot index at all times.[115]

That's what I mean by thinking about things from a first-principles standpoint. If I had analyzed it by analogy and said, "What are all other rocket companies doing? What do their rockets cost? What historically have other rockets cost?" That is reasoning by analogy, but *it really doesn't illustrate what the true potential is*.[116]

> The first-principles approach is a good way to figure out counterintuitive solutions. It was a helpful thing to learn.[117]

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THINKING IN LIMITS

Another good physics tool is thinking about things "in the limit." Take a particular idea and imagine scaling it to a very large or very small number. How do things change?[118]

The Boring Company is a great example. A common criticism of the idea of tunnels is: The tunnel will get used up and there will still be traffic congestion. They don't realize there's no real limit to *how many* levels of tunnel you can have. You can go much farther deep underground than you can build up. The deepest mines are much deeper than the tallest buildings are tall.

Ever notice that cities are built in 3D, but roads are only built in 2D? You could build roads in 3D by building tunnels under cities.[119] You can alleviate any amount of urban congestion with a 3D tunnel network.[120]

But, it's difficult and expensive to dig tunnels the way they're currently done. The LA subway extension cost roughly a billion dollars per mile to build.[121]

> If you do just two things, you can get approximately an order of magnitude improvement in tunneling.
>
> But I think you can even go beyond that.[122]

We need at least a tenfold improvement in the cost per mile of tunneling. The first thing to do is to cut the tunnel diameter. According to current regulations, a single road-lane tunnel has to be twenty-six to twenty-eight feet in diameter. But if you shrink that diameter to twelve feet, the area goes down by a factor of four. This is a huge improvement, because the cost of tunneling scales with the area. That's almost a half-order of magnitude (4–5x) improvement in cost per mile right there.[123]

Currently, tunneling machines work for half the time, then stop. The other half of the time is putting in reinforcements for the tunnel wall. If you design the machine instead to do continuous tunneling *and* reinforcing, that will give you a factor of two improvement. Combine those and that's a factor of eight.[124]

These machines are also far from being at their power or thermal limits, so we should jack up the power to the machine. We can get at least another factor of two there, maybe even four or five. Currently, we're not even close to the limits of tunneling technology.[125]

Q: How does thinking in limits apply to manufacturing?

Let's say we're trying to figure out why a part or product is expensive. Is it because of something foolish, or just because production volume is too low?[126]

Ask, "If our volume was a million units per year, would it still be expensive?" If it's still expensive at a million units a year, then volume is not the reason why your part is expensive. Maybe something's wrong with the design. Maybe you can change the part to something not fundamentally expensive. That's thinking about things to the limit.[127]

If you are really good at manufacturing *and* producing at a high volume, you can make anything for a cost that asymptotically approaches the raw material value of the components, plus any intellectual property you need to license. It's a hard thing to do, but it is possible.[128]

Q: How does thinking in the limit apply to design?

When designing a product, people often start designing with the tools, parts, and methods they are familiar with. That's their default. That will lead to a product that can be made with those tools and methods, but it is unlikely to be the perfect product.[129]

The other way to think is to imagine the platonic ideal of the perfect product or technology. What is the perfect arrangement of atoms that would

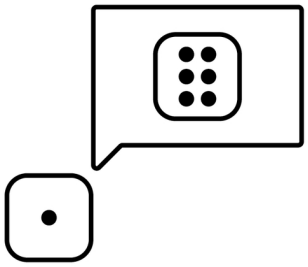
be the best possible product? Now try to figure out how to get the atoms in that shape.[130]

Think through things in both directions. What can we build with the tools that we have? But also, what does the "theoretically perfect" product look like?[131]

The idea of the "theoretically perfect" product is going to be a moving target because as you learn more, the definition for that perfect product will change. You don't actually know what the perfect product is, but you can approximate a *more* perfect product. Then ask, "What tools, methods, or materials do we need to create to get the atoms in that shape?" People rarely think this way. But thinking in limits is a powerful tool.[132]

> Impossible is a strong word. It's just a strong word. I approach things from a physics standpoint, and the word impossible is more or less banned in physics.[133]

At SpaceX especially, we often consider what's possible within the absurd. If my team says something is impossible, I try to open their minds to new potential solutions by asking, "What would it take?"[134]



IT IS OK TO BE WRONG, JUST DON'T
BE CONFIDENT AND WRONG.

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ASPIRE TO BE LESS WRONG

The mental tools of physics are powerful. They tell us to assume we're wrong and that our goal is to be less wrong. Aspire to be less wrong. I don't think you're going to succeed every day in being less wrong. But if you can succeed in being less wrong most of the time, you're doing great.[135]

> It's OK to be wrong. Just don't be confident and wrong.[136]

Aspirationally, you want to believe things proportionate to the evidence. Not inversely proportional to the evidence.[137]

> Most people can learn a lot more than they think they can.
>
> They sell themselves short by not trying.[138]

Read books, because the data rate of reading is much greater than when somebody is speaking. What's the output rate of speech? A couple hundred bits per second, maybe a few thousand per second if you're going full tilt. You can get several times that by reading. The main reason I didn't go to lectures in college was because the data rate was too slow.[139]

> I encourage you to read a lot of books. Just read. Try to ingest as much information as you can.

I was always really interested in reading. When I was a kid, I read everything I could get my hands on.[140] Around nine or ten I ran out of things to read in our house, so in desperation I read the encyclopedia—which turned out to be a good idea. I found all sorts of things I didn't even know existed—a lot, obviously.

I'd recommend everyone read or skim through the condensed version of the *Encyclopedia Britannica*. You can always skip subjects. If you read a few paragraphs and know you're not interested, just jump to the next one.[141]

Develop good general knowledge, so you at least have a rough "lay of the land" of the full knowledge landscape. Read a broad range of material. How can you know what you're really interested in if you're not at least doing a broad, light exploration of the knowledge landscape?[142]

As a kid, I played historical strategy video games, like *Civilization*. This shows you how civilization formed. Through the technology tree, you invent different things. You'd invent literacy, democracy, and gunpowder. You start to realize, "Oh wow, there are stages to technology. You can't have democracy without creating literacy." There are stages of technology and development of ideas. That's a helpful framework.[143]

It is important to view knowledge as a semantic tree. Make sure you understand the fundamental principles (the trunk and big branches) before you get into the leaves (the details), then there is something for them to hang on to.[144]

Some ideas come from reading about a sad trend in technology. When I read about the Concorde being retired, I was like, "Jeez, we don't even have supersonic transport anymore. That's terrible." I never had a chance to fly on the Concorde. That seemed like a terrible thing, so I started reading about it.

I learned the Concorde was designed back in the sixties. Aerodynamics have improved a great deal since then, with computational fluid dynamics. Engine efficiencies have improved massively. Even if you just change the engines on the Concorde, you could double the range or thereabouts. I

thought, *Well, what if we could figure out an efficient design that could make it economically competitive to have a supersonic aircraft?* I started looking into it more and did the math on all of it.[145]

You can get super efficient and super fast when electric aircraft have vertical takeoff and landing, and go supersonic. We could make a breakthrough aircraft several generations beyond what currently exists.[146]

Diving into SpaceX and Tesla, I had to learn how to make hardware. I'd never seen a CNC machine or laid out carbon fiber. I didn't know any of those things, but if you read books and talk to experts, you can pick them up quickly.[147] I started going to the Palo Alto public library to read about rocket engineering and started calling experts, asking to borrow their old engine manuals.[148]

Most people self-limit their ability to learn. It's pretty straightforward—just read books and talk to people.[149]

The physics background is helpful as the foundation. But in rocketry I am self-taught, meaning I don't have an aerospace degree. I just read a lot of books and talked to a lot of people.[150]

> Talk to people from different walks of life, in different industries, professions, and skills. Try to learn as much as possible. Search for meaning.[151]

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